

Continuous Distillation

Pre-Lab Plan
Unit Operations Lab 3
3/04/2026

Group #1, Tuesday 1PM section

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1. Experimental Objective (5pts)

The laboratory's objective is to investigate the steady-state separation of a binary ethanol-water mixture within a continuous distillation column and how feed position and reflux ratios impact column efficiency. The McCabe-Thiele graphical method will be used to compare concentration data from the top/bottom products against theoretical equilibrium stages to evaluate performance.

2. Experimental

The experiment utilized an Armfield UOP3CC Distillation Column [1], which consists of a distillation apparatus containing a reboiler, condenser, trays for the liquid vapor contact, peristaltic feed pump, and thermocouples at different stages of the column. The column is connected to a console that controls various aspects of the distillation process to conduct the experiment. The console is connected to a computer to collect data from the experiment. Refer to figure 1,2 and table 1 for specific details on **Apparatus**.

Experiment will utilize a mixture of ethanol and water to explore the various scenarios to record the effects of different condenser rates and reboiler rates in a continuous distillation process. Efficiency will be measuring the ethanol-water content using the Snap 41 Densitometer. Refer to table 2 for specific details on **Materials and Supplies**.

2.1 Apparatus (5pts)

The system is a binary continuous distillation process designed to separate ethanol and water in a steady state with a continuous feed. The Armfield UOP3CC Continuous Distillation Column [1] consists of multiple trays within the glass columns, a condenser, and top/bottom receivers. Key equipment includes a feed tank, heat exchanger, vent, cooling, and a reflux valve.

The following table provides a complete list of the apparatus components. Each component is numbered to correlate the location of part in the images of the system. As a disclaimer, the tables used are modified from those in the previous binary batch distillation laboratory. Two figures are included; the first is the schematic from the *Discovering with Armfield: Engineering & Research Equipment UOP3CC Instruction Manual*. The second was taken on the day of the laboratory introduction.

Figure 1: Armfield UOP3CC Continuous Distillation Column Schematic [1] [2]

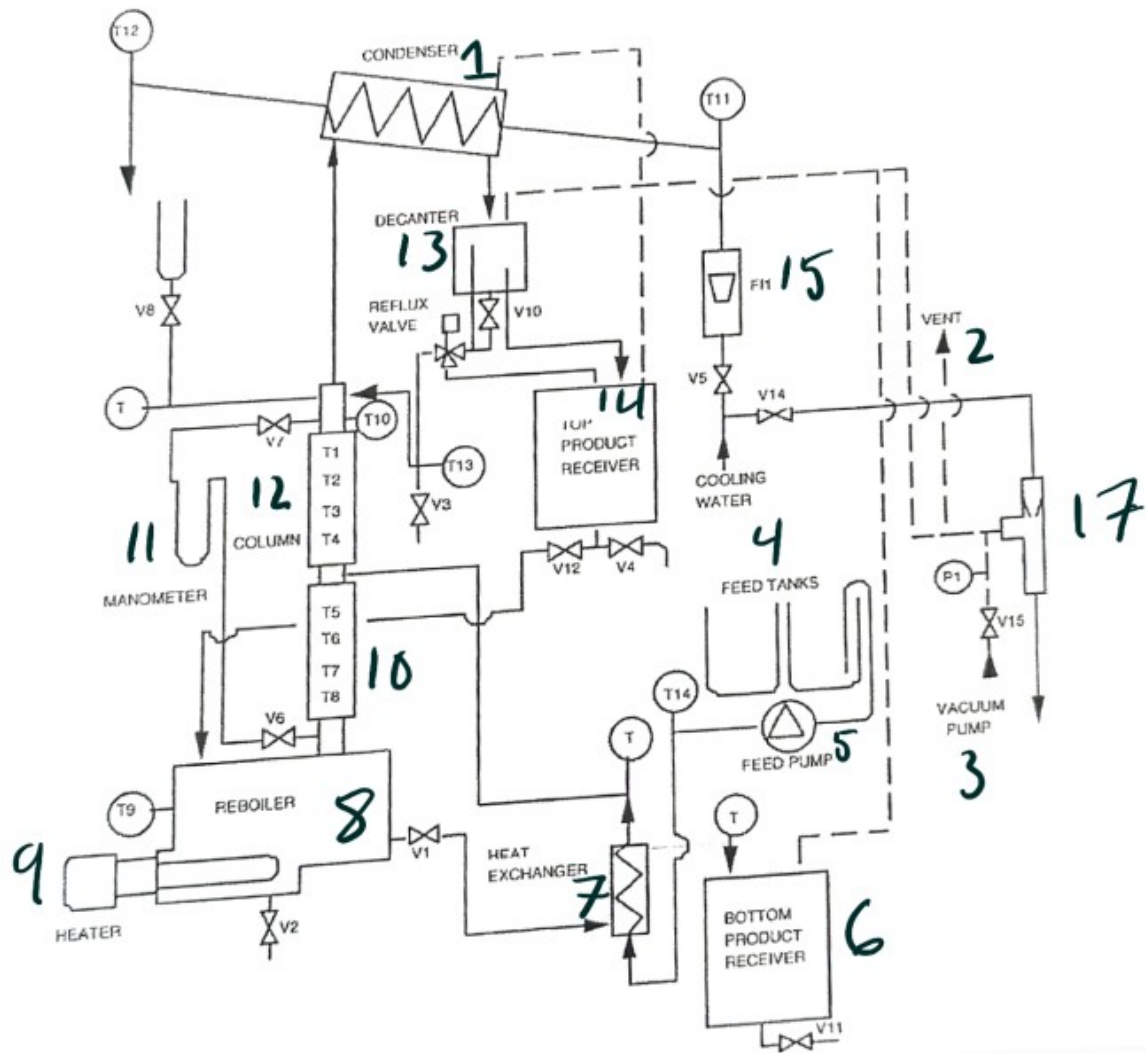


Figure 2: Control Console [1] [2]



Table 1: Apparatus List [2]

Component No.	Component	Description
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1	Condenser	Cools and condenses the water-ethanol solution into liquid distillate.
2	Vent	Releases gas safely from the system.
3	Vacuum Pump	Lowers column pressure below atmospheric to aid in separation capability and efficiency.
4	Feed tank	Stores the ethanol and water before being sent to the column.
5	Peristaltic Feed Pump	Delivers the solution from the feed tank to the column at controlled rates. (0 – 300 mL/min)
6	Bottom Product Receiver	Collects the stream exiting the bottom of the column.
7	Heat exchanger	Transfers heat between the streams.
8	Reboiler	Heats bottom mixture to generate vapor to rise through the column.
9	Immersion heater	Supplies controlled heat to the reboiler.
10	Thermocouples (T1-T9)	Monitors the temperatures in the system. There are 14 in total. We will use T1-T9. On the schematic they are indicated with the letter T and their associated number.
11	Manometer	Measures pressure drop in the column, monitoring the vapor flow and performance.
12	Tray glass column	Allows for multiple stages where vapor and liquid meet to induce the separation of

		ethanol and water.
13	Decanter	Separates any possible immiscible liquid phases of water and ethanol.
14	Top Product Receiver	Collect the condensed distillate.
15	Flow Meter	Controls cooling water used in distillation column.
16	Control Console	Manages the various parameters for the system (Reflux control, Reboiler power, feed pump speed, column, and process temperatures).
17	External Heat exchanger	Uses cooling water to cool various streams around the system.

Table 2: Apparatus Valves List [2]

Valve	Location in Apparatus	Function
V1	Bottom product outlet	Allows the bottom product to exit the column through the heat exchanger to the bottom product receiver.
V2	Reboiler outlet	Controls flow leaving the reboiler toward the heat exchanger and bottom product receiver.
V3	Distillate sampling line	Used to collect overhead samples for composition measurements.
V4	Top product receiver line	Regulates flow into the top product receiver.
V5	Cooling water inlet	Controls cooling water entering the condenser.
V6	Manometer (rectifying section)	Used to measure pressure drop in the upper section of the column.
V7	Manometer (stripping section)	Used to measure pressure drop in the lower section of the column.

V8	Column internal line	Helps regulate internal column liquid/vapor flow (part of reflux routing).
V9	Reflux line	Controls condensed distillate returning to the column as reflux.
V10	Reflux pipe	Allows condensed liquid to return to the column during total reflux operation.
V11	Bottom product receiver outlet	Controls discharge from the bottom receiver.
V12	Distillate receiver control valve	Helps regulate distillate flow into the receiver.
V13	Manometer measurement port	Used for pressure drop monitoring in the column.
V14	Cooling water line	Controls cooling water circulation to the condenser/heat exchanger system.
V15	Vacuum pump line	Regulates connection between the system and the vacuum pump.

2.2 Materials and Supplies (5pts)

Table 3: Materials and Supplies List [2]

Material/Supplies	Description	Material or Supply
200-Proof Ethanol	Solute that will be separated from water.	Material
Deionized Water	Solvent that will be separated from ethanol.	Material
Flask	Used to contain top/bottoms samples from the experiment.	Supply
Funnel	Helps the transfer of solutions to the reboiler and feed tank.	Supply
Test Tubes	Used to contain the distillate and bottoms product.	Supply

Ice/Ice Water	Provides cooling for the samples taken from the column.	Supply
Thermal Gloves	Protects hands when handling hot materials and taking samples from the column.	Supply
Snap 41 Densitometer	Measures the ethanol concentration to evaluate column performance.	Supply
Cooling Water	Circulated through the condenser to produce the distillate/reflux.	Supply
Stopwatch	Measures the time required to collect the distillate and condensate.	Supply
Parafilm	Used to cover samples to prevent evaporation of liquid content.	Supply
Lab Tape	Used to label the beakers that will contain the top and bottoms sample.	Supply

NOTE: This was also modified from the previous distillation laboratory [2].

2.3 Experimental Plan (10pts)

SET-UP:

1. Apply power to the control panel.
2. Begin by making a calibration curve for the peristaltic feed pump. Do this by first disconnecting the tubing from the pump apparatus. Using a stopwatch and a graduated cylinder, determine how long it takes to reach a certain volume of DI water. Repeat this process for ten different pump speeds. Graph the data.
 - a. The pump will need to be turned on by using the control panel.
 - b. DI water will need to be loaded into the feed reservoir.
3. Next, prepare the reboiler and feed solutions
 - a. Reboiler – mix 1.27 L of ethanol in 9.0 L of water

- b. Feed – mix 0.76 L of ethanol in 2.4 L of water
 - c. **NOTE:** Be sure to measure out liquids in separate glassware and to mix the ethanol into the water
- 4. Close all valves except for V10 (on the reflux pipe).
- 5. Fill the reboiler and feed reservoirs with their respective solutions by using a funnel. Be sure to attach the blue caps afterwards for concealment.
- 6. Reconfigure the tubing back into the pump and attach the rest of the feed tubing to the middle of the distillation column (between trays 4 & 5).
- 7. Configure the equipment to begin operation with total reflux. Do this by switching off the reflux ratio timer on the console

EXPERIMENT 1 – STEADY STATE DISTILLATION:

1. Start by setting the temperature selector switch to T9 (the reboiler), open V5 (cooling water to condenser) and make the flowrate approximately 3 L/min (on F11).
2. Ensure that the column will operate in total reflux by checking that the switch for the reflux ratio timer is in the off position.
3. Set the reboiler power by first ensuring that the power controller knob is turned fully counterclockwise. Then flip the switch that applies power to the heating element. Now turn the power knob clockwise until a reading of 1.5 kW is obtained on the digital wattmeter.
 - a. In this area of the control panel, a red light should illuminate indicating that the heating element is on.
4. Wait until vapor appears within the column and enters the distillate receiver as condensate. After some build-up, the condensate will begin to overflow to the reflux regulator valve.
 - a. Vapor formation can be monitored by switching the temperature selector through options T1-T8.
 - b. Total reflux should be in effect at this point.
5. Turn the reboiler power down to 0.75 kW and allow the system to stabilize while in total reflux.
 - a. This takes anywhere from 15-30 minutes.
6. Once the system is stabilized; begin feed flow (by turning on the pump), bottoms flow (opening V1) and change the reflux to a ratio of 5:1.
 - a. This reflux ratio means by five seconds back to the column (C-) and one second to the top product receiver (C+).
 - b. To apply the reflux ratio, switch the reflux valve on by using the control panel. There should be an audible clicking sound, this means that the reflux is working.
 - c. Condensate should be flowing to the top product receiver at this point.

- d. A feed flowrate of 30-35 mL/min is advisable.
7. After feeding around 400 mL (~15 minutes), take a sample of the overhead (V3) and bottoms (V2). This should be done at the same time.
 - a. **CAUTION:** liquid will be hot!
 - b. **NOTE:** cover samples waiting to be analyzed by the densitometer with parafilm.
 - c. Take two more samples five minutes apart.
 - d. Be sure to record the temperatures in the distillation column (T1-T9).
 - e. Record the ethanol concentration in the samples using the densitometer (SNAP 41).
8. After concentration values have been recorded, turn off the feed pump, the reflux switch, the reboiler power, and close V1.

EXPERIMENT 2 – VARYING THE FEED POSITION:

1. Start by disconnecting the feed tube from the column midpoint and reconnect the plug.
 - a. **CAUTION:** tube and column will be hot; liquid may leak out as well. Wear heat resistant gloves if possible.
2. Remove the plug from the position above the top plate and connect the feed tubing there.
3. Configure the column to operate at total reflux using the same steps described in experiment 1.
4. Turn the reboiler power to 0.75 kW and allow the system to stabilize for 15-30 minutes.
 - a. This process should be quicker than before since the column is still warm from before.
5. After stabilization, set the reflux ratio back to 5:1 and switch the reflux valve to on using the control panel.
 - a. Listen for an audible click to verify that reflux is working.
 - b. There should also be condensate flowing the top product receiver.
6. At this point, V1 can be opened to allow for the bottoms to flow to the receiver, the feed pump can be turned on.
 - a. A feed flowrate of 30-35 mL/min is advisable.
7. After feeding 400 mL (~15 minutes), take a sample from the overheads (V3) and the bottoms (V2). Do this at the same time.
 - a. Take two more samples five minutes apart.
 - b. Record the temperatures in the column (T1-T9).
 - c. **CAUTION:** Liquid will be hot!
8. Measure and record the concentration of ethanol in each sample using the densitometer (SNAP 41).

- a. **NOTE:** cover samples waiting to be measured with parafilm.
9. After samples have been measured and recorded; turn off the feed pump, the reflux switch, and the reboiler power. Also, close V1.

INDEPENDENT/CONTROLLED VARIABLES: reboiler/condenser duty, reflux ratio, feed position, feed flowrate, feed composition, and condenser cooling rate.

DEPENDENT VARIABLES: bottoms/overhead composition, tray temperature/pressure, tray composition, Murphee plate efficiency, and ΔP in the column.

POST-LAB PLAN:

1. For the unsteady state portions of this lab, the recorded temperatures as a function of time will be plotted.
 - a. These temperatures will be related to the column approaching steady state.
2. An ideal McCabe-Thiele analysis, using equilibrium data from the two feed positions, will be performed.
 - a. Results from this analysis will be compared with the experimental results. They will also be thoroughly explained.
 - b. An energy balance will also be made to help understand where energy loss might've occurred during the process.
3. The column efficiency for both feed positions will be calculated as well.

Project Safety Evaluation (10pts)

Project Title: Continuous Distillation

Date: 3/04/26

	<i>Yes/No</i>	
Potential electrical hazards?	Yes	If yes, identify conditions: Conditions come from the control console and reboiler as the reboiler operates at a high power and there is a risk of shock is fluid spills touch the control console
Potential mechanical hazards?	Yes	If yes, identify conditions: Conditions come from the peristaltic feed pump as there are potential pinch points that are created when using the tubing into the rotor and clamp.
Potential pressure hazards?	Yes	If yes, identify conditions: Conditions come from the reboiler and the vent pipe as pressure buildup can happen if the reboiler cap is not placed firmly or if the cooling water doesn't condense vapor
Potential temperature hazards?	Yes	If yes, identify conditions: Conditions come from the reboiler, column trays and product samples as the column trays warm up as the vapor rises through it and the product samples taken from valves V2 and V3 will be hot.
Hazardous/toxic chemicals involved?	Yes	If yes, hazards involved: Flammable ethanol is an irritant and presents a fire hazard when heated.
Gloves required?	Yes	If yes, specify type: Insulated thermal gloves are needed when collecting hot samples from valves V2 and V3.
Other hazards	Yes	If yes, specify conditions: Inhaling ethanol vapors due to a leak or crack presents a health hazard. Inhalation and/or skin contact is possible if samples aren't treated with care.
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Ethanol: (CAS# 64-17-5) Deionized (DI) water: (CAS#7732-18-5)
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	~100 °C	Location: Reboiler (T9) Hazards: Burn hazards from boiling liquid. Precautions: Use insulated thermal gloves and do not open cap

		while unit is running.
Pressure (psia)	1.5 kW	Location: Digital wattmeter Hazards: Electrical fire / over heating Precautions: Reduce the power to 0.75 kW as the stabilization begins and watch the temperatures of the trays (T1-T8).
Safety Controls	yes/no	If yes, Provide description
Reflux Pipe (V10)	Yes	<i>This makes sure the condensed vapor goes back to the column during the total reflux to stabilize the system before sampling begins.</i>
Cooling Water Flow (F11)	Yes	<i>The monitors the flow rate of the condenser and must be kept at around 3 L/min to stop the vapor escaping.</i>
Vent Pipe	Yes	<i>This is placed at the side of the unit allowing the uncondensed vapor to emerge if the cooling water flow isn't enough.</i>

REFERENCES:

1. Armfield. *Continuous Distillation: ChE 382 Unit Operations Laboratory Manual*; Spring 2024 v1.
2. Silva, J.; Laforest, J.; Abuzir, A.; Dilosa, C. *Binary Batch Distillation: Pre-Lab Plan, Unit Operations Lab #2*; ChE 382 Unit Operations Laboratory, Tuesday 1 PM Section, September 16, 2025.

SAFETY STATEMENT:

I have read relevant background material for the Unit Operations Laboratory entitled: “Continuous Distillation” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance with standards and acceptable safety practices and will conduct all my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

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Date: 04 March 2026

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